CMPG 325 – Individual Semester Project

Milestone 1

Client Design Review

Network Design Project

Katlego Adult Learning Centre

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Milestone 1 Submission Package

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1 Introduction

This document clearly sets out the client requirements analysis, network topology, and IP addressing plan for the Katlego Adult Learning Centre, located in Rustenburg, North West Province, South Africa. It establishes the operational context and functional requirements that the network design is built upon, before presenting the physical and logical topology and the full IP addressing scheme.

The purpose of this documentation is to translate the client's operational backround or context into concrete, justifiable networking requirements, and to demonstrate how those requirements are addressed through network segmentation, topology design, and IP addressing, ahead of implementation in Cisco Packet Tracer.

2 Client Background

Katlego Adult Learning Centre is a community-based adult education facility operating in Rustenburg, a city in the North West Province best known as the centre of South Africa's platinum mining industry. Rustenburg's population includes a significant number of mineworkers and their families, many of whom historically had limited access to formal schooling due to work commitments or the broader legacy of unequal access to education in South Africa.

Adult learning centres such as Katlego exist specifically to address this gap, offering adults a second opportunity to gain literacy, numeracy, and basic digital skills. The client is best understood not as a large corporate entity but as a modest, single site community institution operating with a limited budget and a small, focused staff and learner base.

This background directly informs the scale and design philosophy of the network proposed across, the solution must be realistic, cost-appropriate, and sized for a small institution, while still demonstrating sound networking principles.

3 Project Scope

The scope of this project, as defined in the client brief (CMPG325-2026-025), is to design and simulate a computer network for Katlego Adult Learning Centre that:

- Provides appropriate connectivity and network services across the client's facility.
- Makes use of the assigned addressing block `192.168.22.0/24`.
- Implements the assigned technical challenge: **Port Security (switchport access control)**.
- Accommodates the stated design constraint regarding restricted server room access.
- Accommodates Change Request CR11 (adoption of VoIP handsets) within the existing design, without introducing additional scope beyond what is required.

4 Facility Overview

Based on the nature and scale of the client organisation, the facility is modelled as a small, single storey building comprising seven functional spaces. This layout reflects a realistic footprint for a community adult learning centre and provides sufficient variety to demonstrate sound network segmentation without introducing unnecessary complexity.

Table 1: Functional spaces within the Katlego Adult Learning Centre facility

Room	Function
Classroom A	General instruction space for literacy/numeracy classes
Classroom B	General instruction space for a second subject stream
Computer Lab	Digital literacy training; highest device density on site
Admin Office	Registration, finance, and centre management
Reception	First point of contact for learners and visitors
Library / Resource Room	Shared self-study computer access for learners
Server Room	Houses core network infrastructure; access-restricted

Design Rationale: I modelled seven rooms rather than a larger or more elaborate floor plan because it is the smallest layout that still lets me demonstrate distinct traffic types which are, classroom, lab, admin, voice, and restricted infrastructure without padding the design with spaces that would not exist in a real facility this size.

5 Device Inventory

The following device inventory was developed to reflect realistic usage patterns for a small adult education centre. Class sizes were kept modest, consistent with the way adult basic education is typically delivered around learners' work and family commitments, and device counts were deliberately kept lean to avoid over-provisioning a client with a limited operating budget.

Table 2: Device inventory by room

Room	Devices	Count
Classroom A	Teacher PC	1
Classroom B	Teacher PC	1
Computer Lab	Learner PCs and the Instructor PC	11
Admin Office	Staff PCs and a shared printer	3 PCs, 1 printer
Reception	Front-desk PC and a VoIP handset	1 PC, 1 phone
Library / Resource Room	Shared learner PCs	2
Server Room	Core switch, router (infrastructure only)	–

Provisioning Summary. In total, the facility requires provisioning for **20 end-user devices** and **1 VoIP handset**, with the design allowing headroom for additional devices, particularly further VoIP handsets, as anticipated by Change Request CR11.

Design Rationale: I placed only one VoIP handset at Reception for this milestone, rather than rolling handsets out across every room, because CR11 introduces voice as a new requirement rather than a full deployment and a single handset is enough to justify a dedicated voice VLAN while leaving clear, deliberate headroom for the client to expand later.

6 Functional Requirements

Based on the client background, facility layout, and device inventory above, the network must satisfy the following functional requirements:

1. **Connectivity:** All end-user devices across all seven rooms must be able to reach appropriate network resources and, where required, communicate with one another.
2. **Segmentation:** Traffic must be logically separated according to function to reflect differing trust levels and traffic types across the organisation.
3. **Scalable addressing:** The IP addressing scheme must make efficient use of the assigned `192.168.22.0/24` block using Variable Length Subnet Masking (VLSM), sized appropriately per segment with reasonable room for growth.
4. **Port-level access control:** In line with the assigned technical challenge, switch ports must be configured with Port Security to prevent unauthorised devices from connecting to the network, particularly given the client's limited on-site IT oversight.
5. **Restricted infrastructure access:** The server room and its associated network infrastructure must be logically isolated, satisfying the constraint that access is limited to authorised IT staff only.
6. **Voice readiness:** The network must accommodate VoIP handsets, as required by CR11, through a dedicated voice segment that avoids contention with regular data traffic.

7 Network Segmentation Overview

To satisfy the functional requirements above, the network is divided into five logical segments (VLANs), grouped by function rather than by physical room alone. This approach keeps the design simple enough to implement and defend confidently, while still reflecting sound, industry-standard segmentation practice.

Table 3: Logical network segmentation by VLAN

VLAN	Segment	Justification
VLAN 10	Data (Classrooms & Library)	Low-density learning-support spaces with similar, light traffic needs
VLAN 20	Data (Computer Lab)	Highest device density on site; isolated to its own broadcast domain
VLAN 30	Data (Admin)	Business or administrative operations, which is different from learner traffic
VLAN 40	Voice	Dedicated segment for VoIP handsets per CR11, avoiding contention with data traffic
VLAN 99	Management/Server	Isolated infrastructure segment, satisfying the restricted-access constraint

8 Physical Topology

The physical topology below shows the actual devices and cabling structure of the network. The Router and Core Switch are located in the Server Room, satisfying the client's access-restriction constraint. Three access switches serve the remaining rooms, grouped to match the VLAN structure: There is one for Classrooms and the Library, one for the Computer Lab, and one for Admin and Reception.

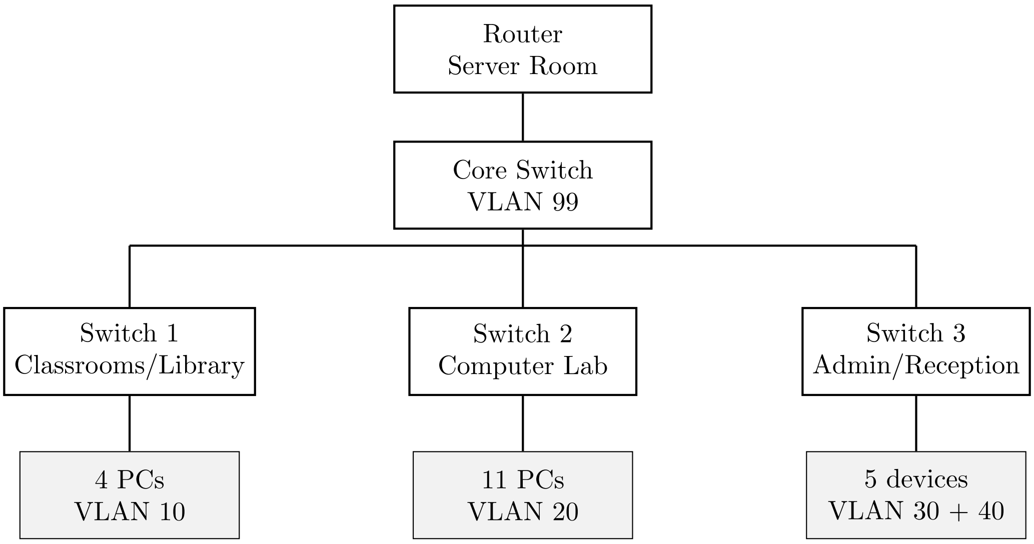

Figure 1: Physical topology of the Katlego Adult Learning Centre network

Cabling summary: Router to Core Switch is a single trunk link. Core Switch to Access Switch links are trunk links carrying tagged VLAN traffic. Access Switch to end-device links are standard access-mode ports, each configured individually with Port Security as required by the assigned technical challenge.

9 Logical Topology

The logical topology below shows how traffic flows between VLANs, independent of physical location. The Router uses sub-interfaces (router-on-a-stick configuration) to route between the five VLANs over a single physical trunk link to the Core Switch.

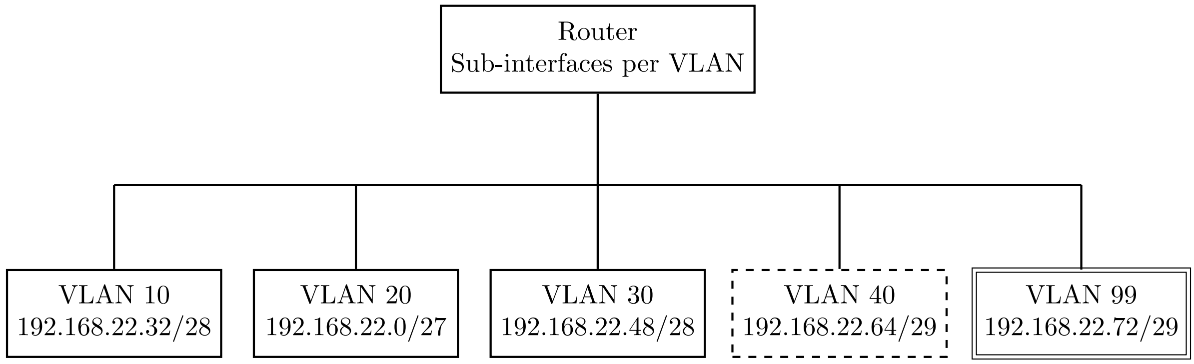

Figure 2: Logical topology showing VLAN-to-subnet mapping via router sub-interfaces

10 IP Addressing Plan

10.1 Addressing Methodology

The client has been assigned a single Class C-equivalent block, `192.168.22.0/24`, providing 256 total addresses (254 usable) before subnetting. Rather than dividing this block into equal fixed-size subnets, Variable Length Subnet Masking (VLSM) was used to size each subnet according to the actual number of devices it must support, plus reasonable headroom for future growth.

This approach was chosen for three reasons:

1. **Efficiency:** The client's segments vary significantly in size, from 30 devices (Computer Lab) to a handful of infrastructure addresses (Management VLAN). A fixed size split would either waste address space on small segments or fail to accommodate larger ones.
2. **Scalability:** Each subnet includes headroom beyond its current device count, allowing the client to add devices, particularly further VoIP handsets under CR11 without requiring the addressing plan to be redesigned.
3. **Demonstrated understanding:** VLSM reflects industry standard addressing practice and directly satisfies the scalable addressing functional requirement.

Subnets were allocated largest-first, a standard VLSM convention that avoids fragmentation of the remaining address space and simplifies later recalculation should further segments be added.

10.2 Address Requirements per Segment

Table 4: Address requirements per VLAN, including growth headroom

VLAN	Devices now	Addresses allocated (including their potential growth)
VLAN 20 – Computer Lab	11	30
VLAN 10 – Classrooms & Library	4	14
VLAN 30 – Admin	5	14
VLAN 40 – Voice	1	6
VLAN 99 – Management	infrastructure only	6

The Computer Lab was allocated the largest subnet as it holds the highest device density on site. The Voice and Management VLANs were deliberately sized small, since they support a single handset and infrastructure devices respectively at present, but each retains a small margin for expansion.

10.3 VLSM Subnet Allocation

Allocating VLAN 20 first, as the largest subnet, and proceeding to progressively smaller subnets avoids address fragmentation. The address range `192.168.22.80` through `192.168.22.254` remains unallocated and is reserved for future expansion, such as additional VLANs or larger than anticipated growth in any existing segment.

Table 5: VLSM subnet allocation for the 192.168.22.0/24 block

VLAN	Network	Mask	Usable range	Broadcast	Gateway
VLAN 20	192.168.22.0	/27	.1 – .30	192.168.22.31	192.168.22.1
VLAN 10	192.168.22.32	/28	.33 – .46	192.168.22.47	192.168.22.33
VLAN 30	192.168.22.48	/28	.49 – .62	192.168.22.63	192.168.22.49
VLAN 40	192.168.22.64	/29	.65 – .70	192.168.22.71	192.168.22.65
VLAN 99	192.168.22.72	/29	.73 – .78	192.168.22.79	192.168.22.73

10.4 Subnet Calculation Reference

For completeness, this section shows the calculation underlying one representative subnet VLAN 10 (Classrooms & Library) to demonstrate the addressing logic applied consistently across the plan.

Requirement: 4 devices currently, sized to 14 usable addresses for growth.

Bits required: $2^n - 2 \geq 14$ requires $n = 4$ host bits, leaving a subnet mask of `/28`.

Resulting subnet: `192.168.22.32/28`

- Network address: `192.168.22.32` (not assignable)
- First usable address (gateway): `192.168.22.33`
- Last usable address: `192.168.22.46`
- Broadcast address: `192.168.22.47` (not assignable)

The same method was applied to each of the remaining VLANs in the table above.

10.5 Mapping to Functional Requirements

Table 6: Mapping of IP addressing decisions to client functional requirements

Functional requirement	Addressed by
Segmentation	Each functional group (learning spaces, lab, admin, voice, infrastructure) receives its own subnet and broadcast domain
Scalable addressing	VLSM sizing with headroom in every subnet; 175 addresses reserved for future use
Restricted infrastructure access	VLAN 99 isolated to its own /29 subnet, separate from all end-user traffic
Voice readiness (CR11)	VLAN 40 provides a dedicated /29 voice subnet, isolated from data traffic